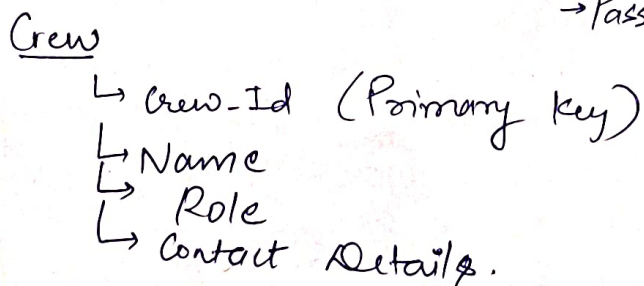
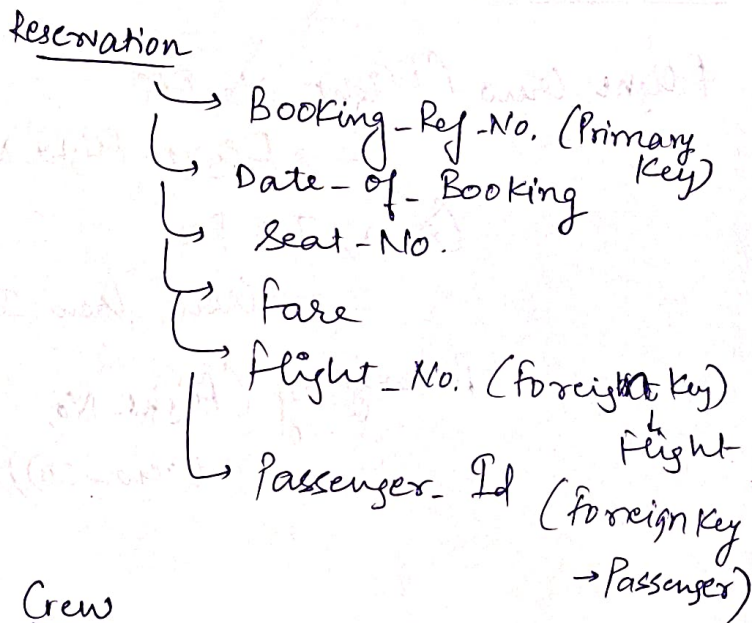
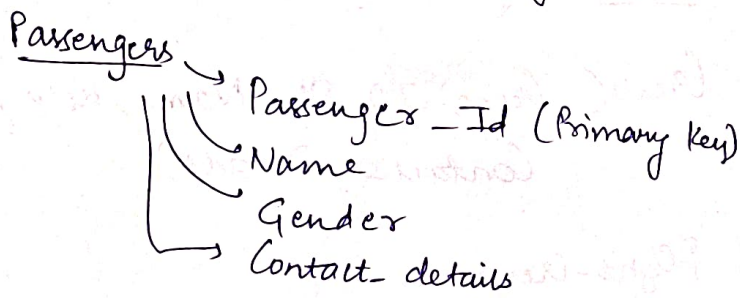
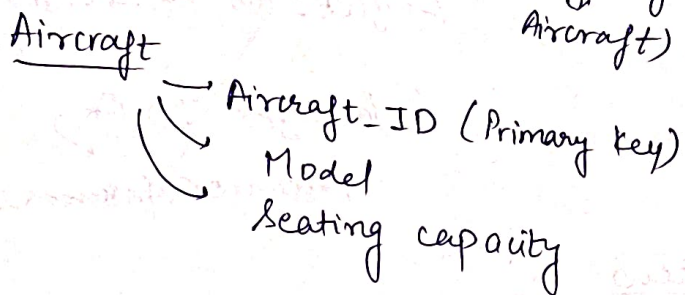
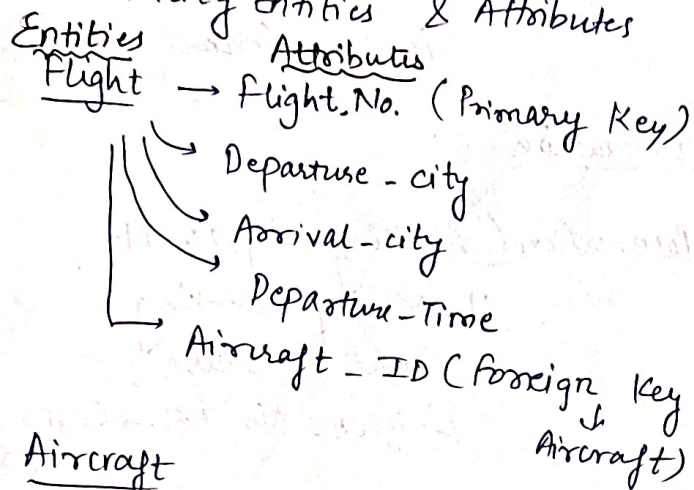
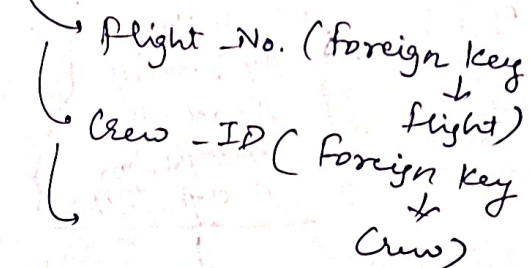


4. Airline Reservation System :-

• Identity Entities & Attributes



Flight-crew (for many - to
 - many between
 Flight and crew)

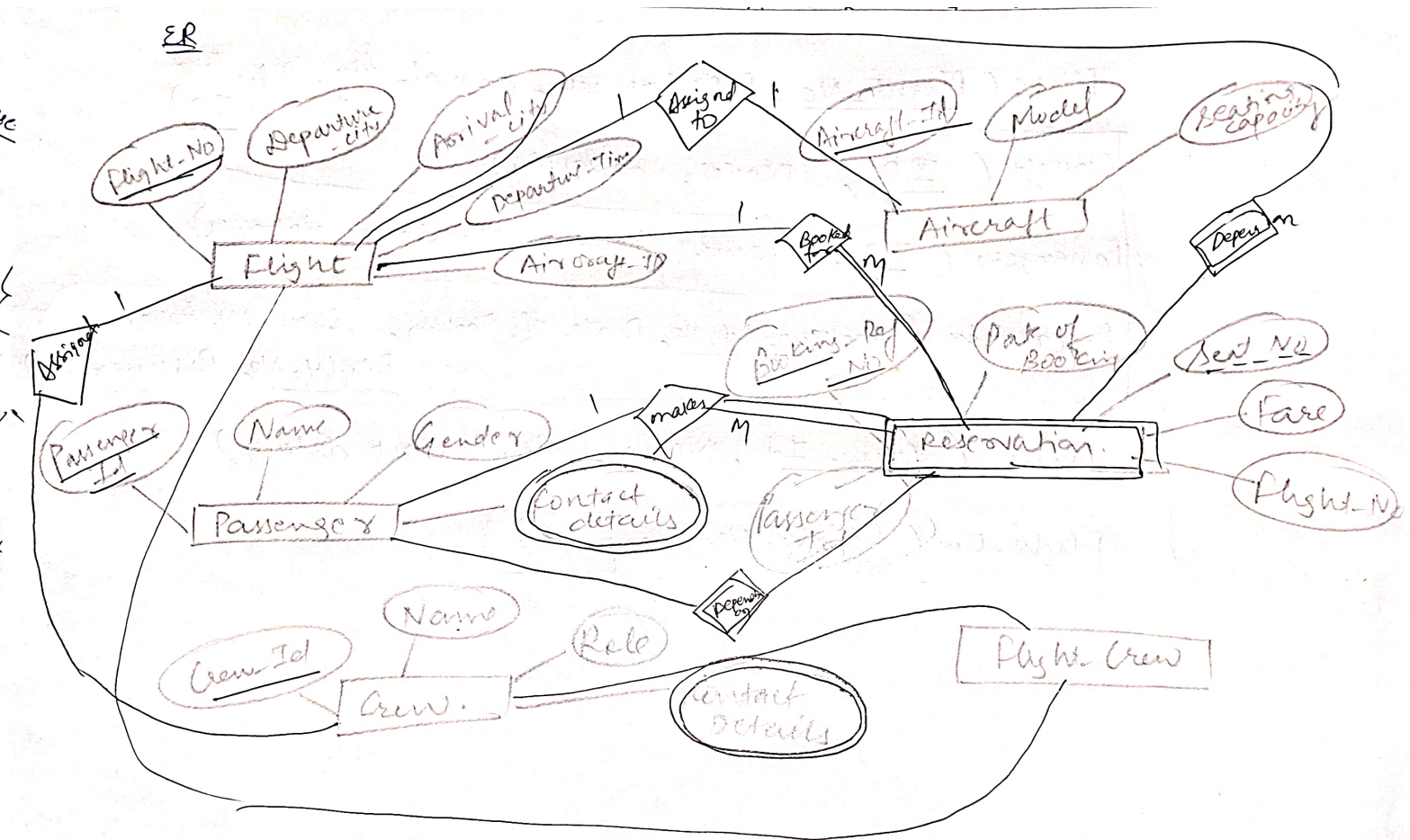


Composite primary key:
 Flight.No + Crew-ID.

• Determine Relationships & cardinalities

Relationship	Description	Cardinality
Flight-Aircraft	each flight ↓ 1 aircraft used many flights	many (Flight) One (Aircraft)
Passenger-Reservation	passenger ↓ many reservations each reservation for 1 passenger	One (passenger) Many (Reservation)
Flight-Reservation	A flight ↓ many reservation each reservation is for one flight	One (flight) Many (Reservation)
Flight-Crew	A flight ↓ many crew members a crew members can work on many flights.	Many Many (Handled by Flight-crew)

ER



Flight (Flight-No, Departure-city, Arrival-city, dep.-Time, ID)

Aircraft (ID, Model, capacity)

Passenger (ID, Name, Gender, Contact details)

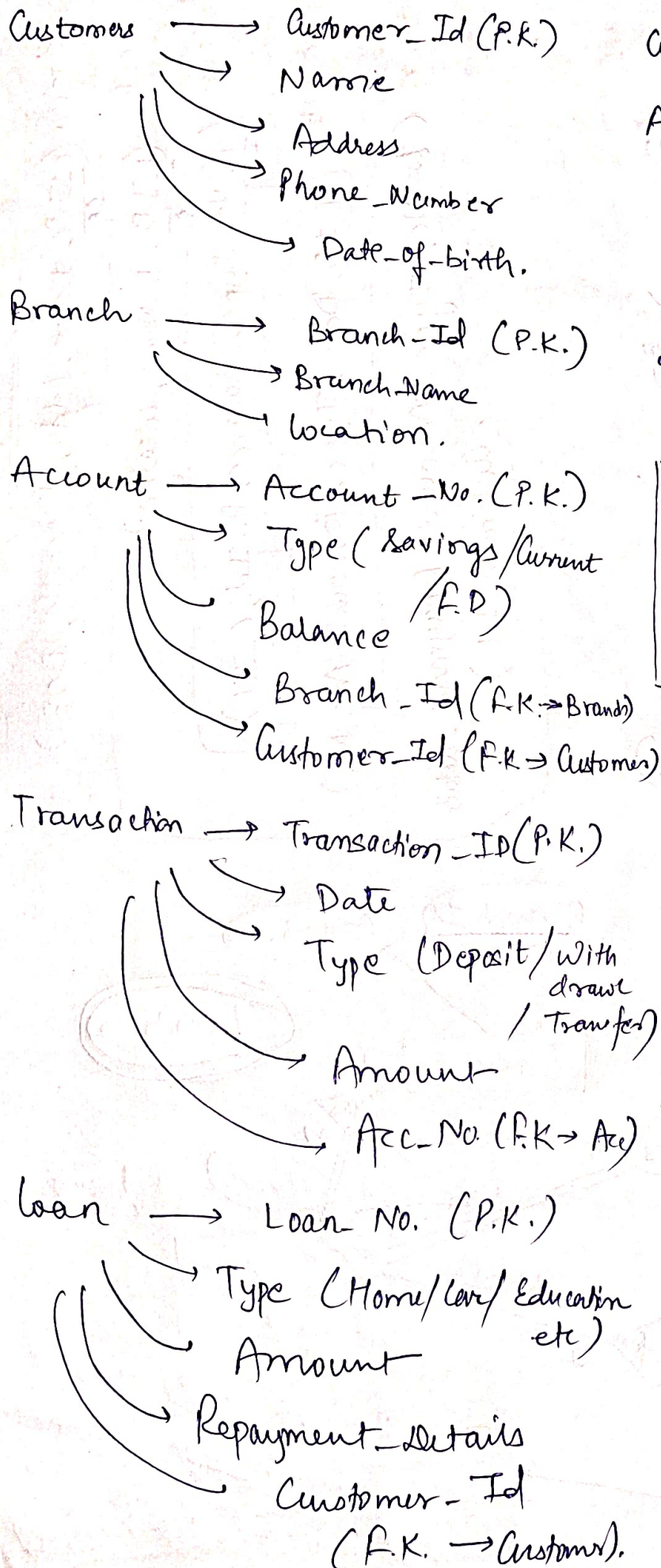
Reservation (Booking-Ref-No, Date-of-Booking, Seat-No, Fare, Flight-No, Passenger-ID)

Crew (Crew-ID, Name, Role, Contact details)

Flight-crew (Flight-No, Crew-ID)

5. Banking Management System:-

Entities & Attributes



Determine Relationships & Cardinalities.

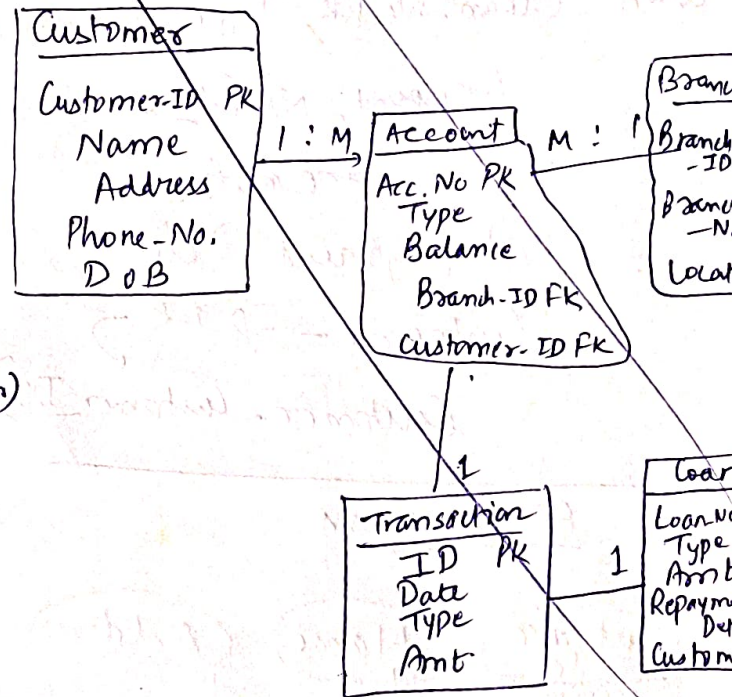
m:many.

Relationships

Cardinality

Customer - Account	1 : M
Account - Transaction	1 : M
Branch - Account	1 : M
Customer - Loan	1 : M.

ER Diagram

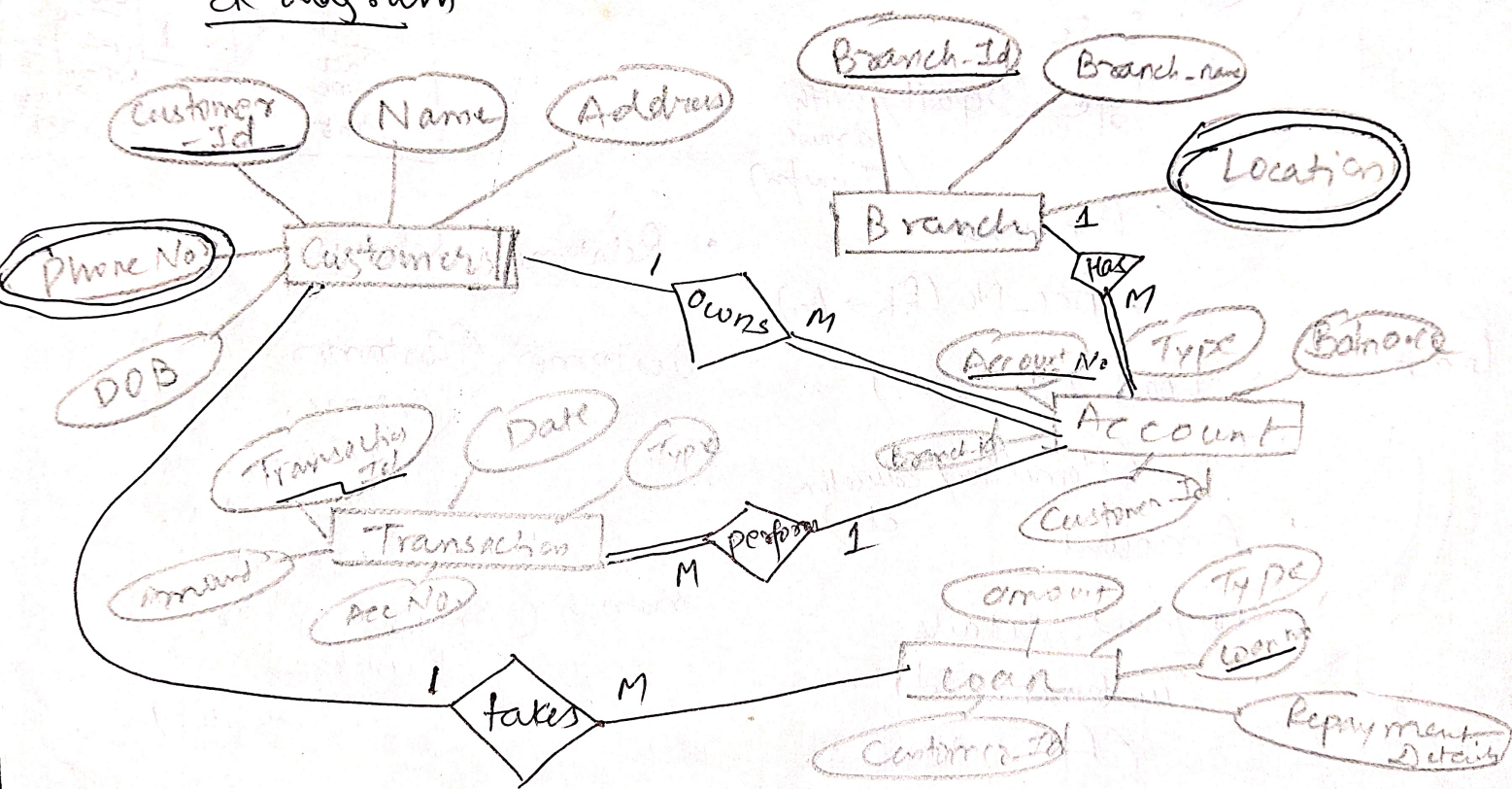


Relational Schema

Customer (Customer-ID PK, Name, Address, Phone-No., DOB)

Branch (Branch-ID PK, Branch-Name, Location)

ER diagram



Relational schema

Customers (C-ID, Name, Addr, Phone-No, DOB)

Branch (B-ID, B-Name, location)

Account (A-No, Type, Balance, B-Id, C-Id)

Transaction (T-ID, Date, Type, Amount, Acc-No.)

Loan (Ln-No, Type, Amount, Repayment_details

, Customer-Id)

12. Music Streaming System:-

- Identity Entities & Attributes.

Entities

Attributes

Artist → Artist ID (P.K.)

Name

Genre.

Album → Album ID (P.K.)

Title

Release year.

Artist ID (F.K.)
of Artist.

Song → Song ID (P.K.)

Title

Duration

Album ID (F.K.)
of Album.

User → User ID (P.K.)

Name

email

Subscription Type
(free/premium)

Payment details

Renewal Date.

Playlist → Playlist ID (P.K.)
Name
User ID (F.K.)
from User.

Playlist-song → Playlist ID (F.K.)
↓
Composite P.K.
(Playlist ID + Song ID)
Song ID (F.K.)
Artist ID (F.K.)

Play-history → History ID (P.K.)
User ID (F.K.)
Song ID (F.K.)
Play Timestamp - can
Last played.

- Relationships & Cardinalities

Relationship

Cardinality

Artist - Album

1 : M

Album - Song

1 : M.

User - playlist

1 : M.

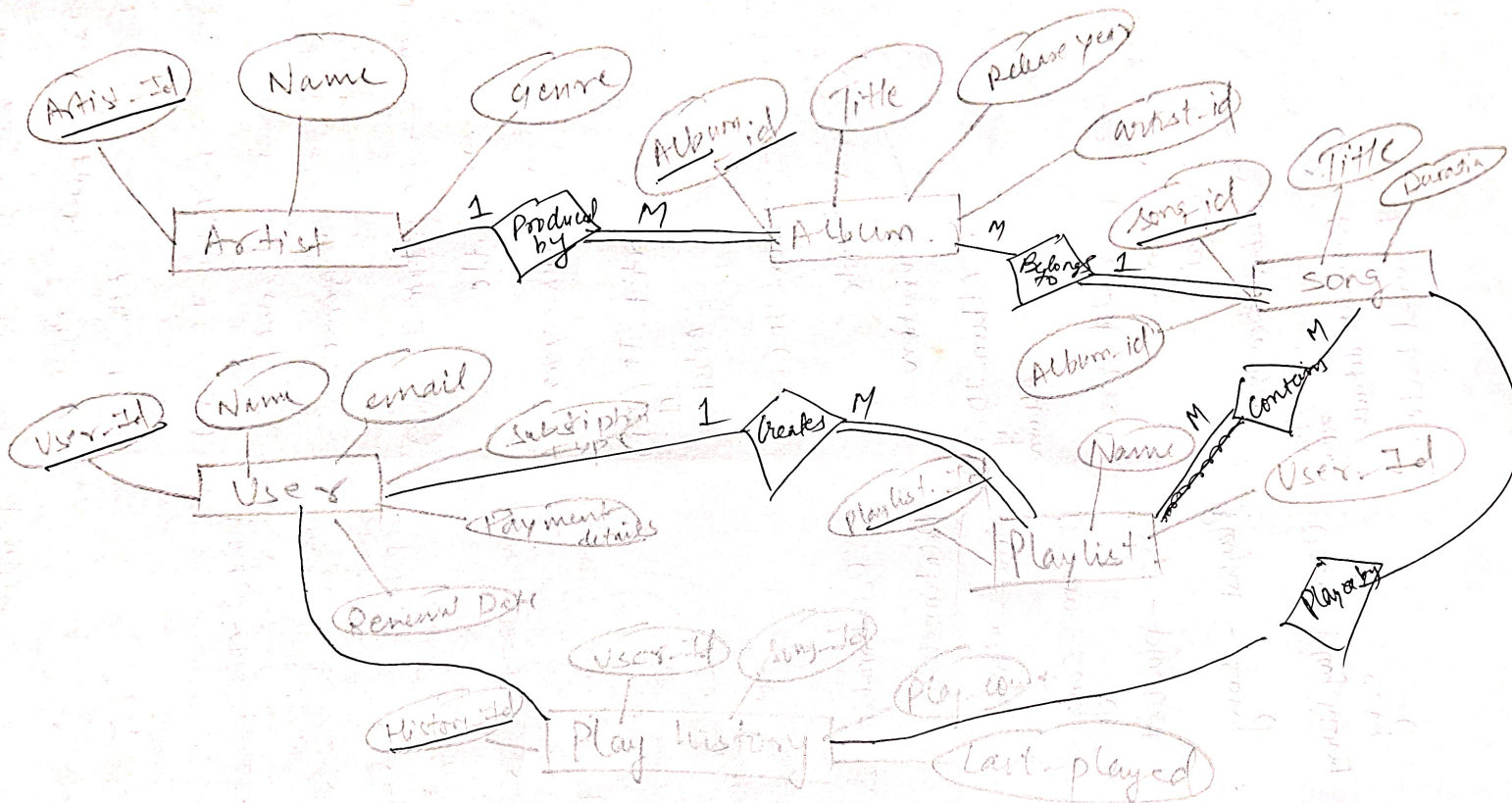
Playlist - Song

M : M.

User - play-history
- Song

M : M.

ER



Relational

Artist (A-ID, Name, Genre)

Album (Album-ID, Title, Release year, Artist-ID)

Song (S-ID, Title, duration, Album-ID)

User (V-ID, Name, email, subscription Type, Payment-details, Renewal Date)

Playlist (Playlist-ID, Name, V-ID)

Play-history (History-ID, V-ID, S-ID, Play-count, last-played)